

One Encoder for Shared Physics

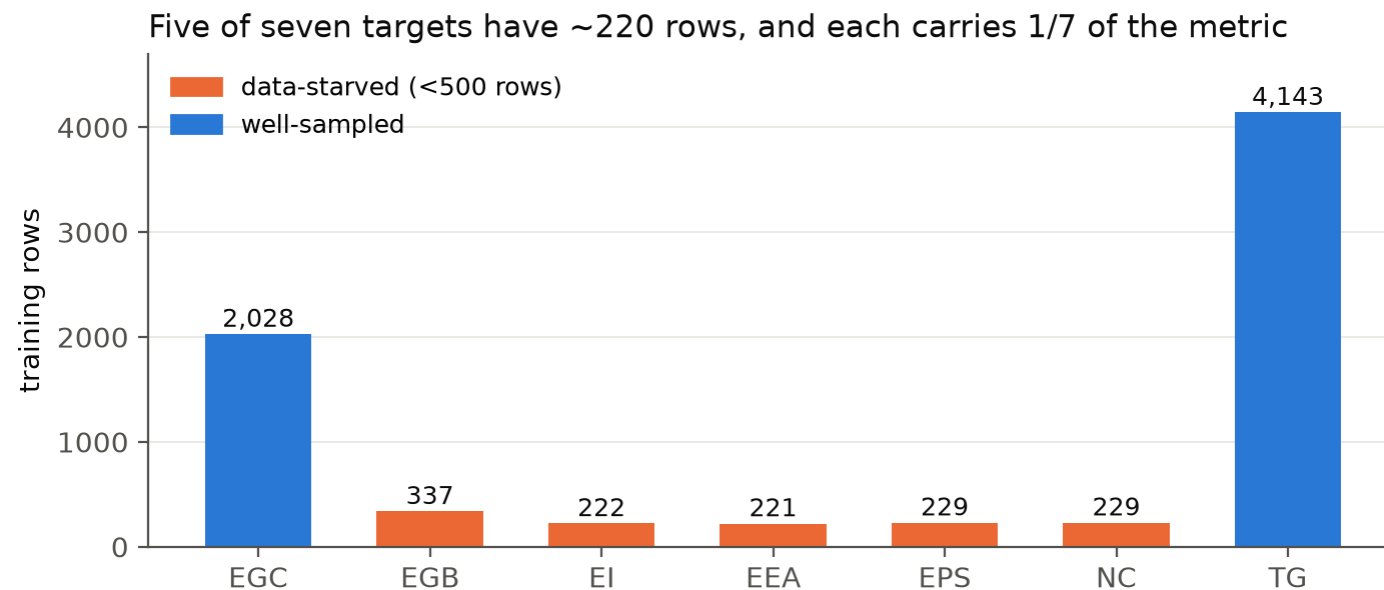
Multi-task learning across correlated polymer properties, with measured representation invariance

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Amalgam Hackathon — Round 3 Finale · Polymer Property Prediction · September 2026

The problem is not accuracy. It is imbalance.

Seven targets, one unweighted mean of per-target R^2 . Five targets have ~220 training rows each.



5 of 7

targets under 500 rows

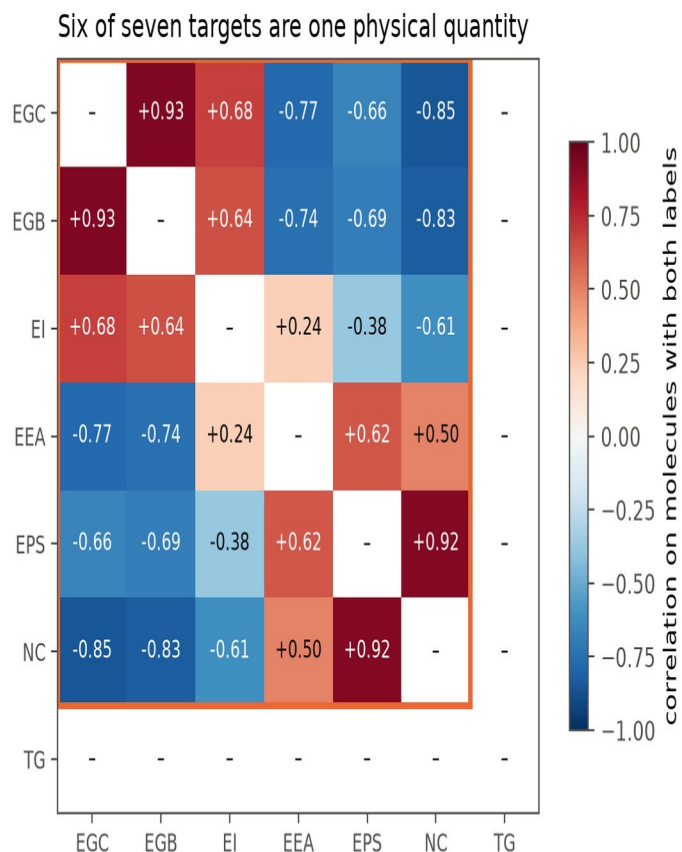
71%

of the metric carried by
14% of the data

A per-target LightGBM reaches only R^2 0.75 on dielectric constant. Fitting these independently is hopeless.

We measured the targets before modelling them

On the 415 molecules carrying more than one label, the structure is unambiguous.



+0.93 chain vs bulk band gap

+0.92 dielectric constant vs refractive index — the textbook $n^2 \approx \epsilon$

-0.85 chain band gap vs refractive index

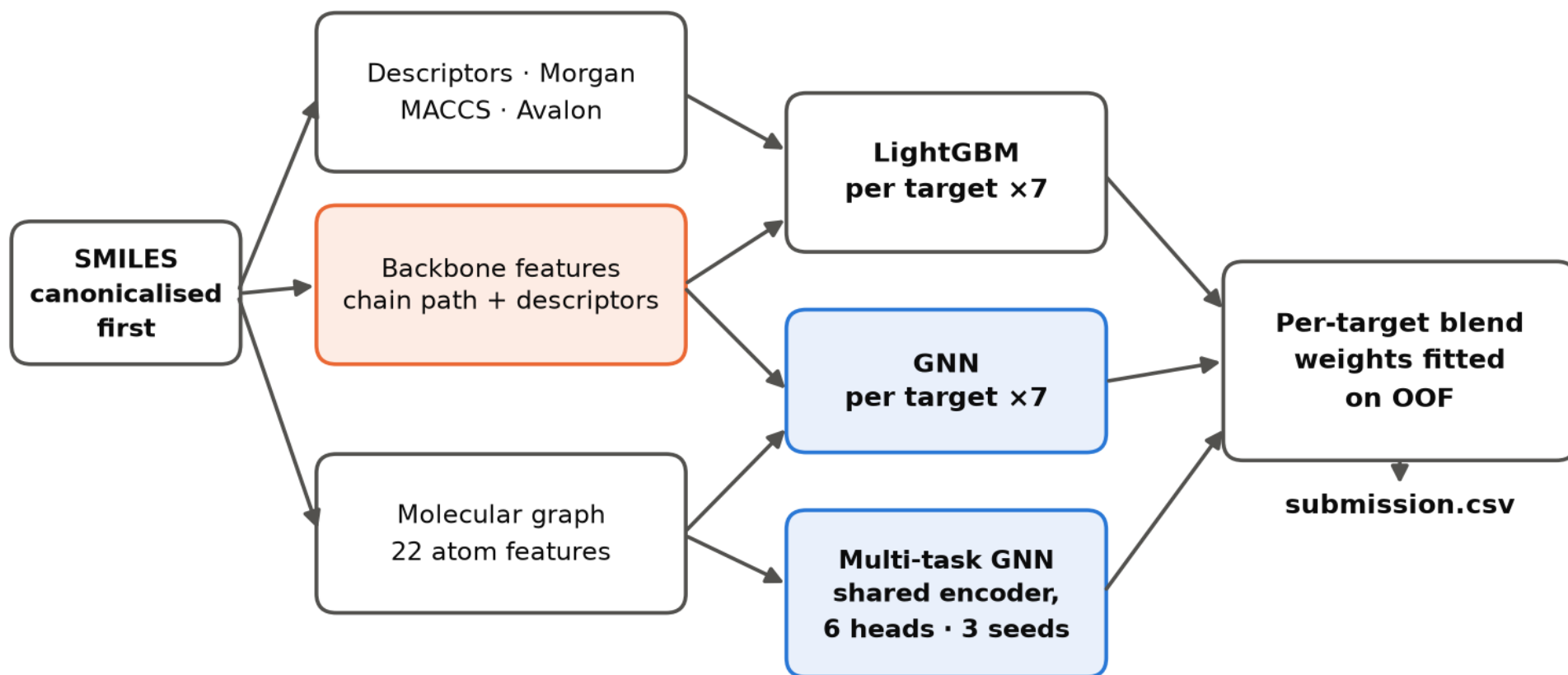
All six are frontier-orbital quantities. Glass transition is thermal and shares no molecule with any of them — so it trains alone.

Orange box: the electronic/optical cluster that shares one encoder. TG row and column are empty because no molecule carries TG and another label.

Architecture

Tree path, graph path, and one encoder shared across the six electronic targets.

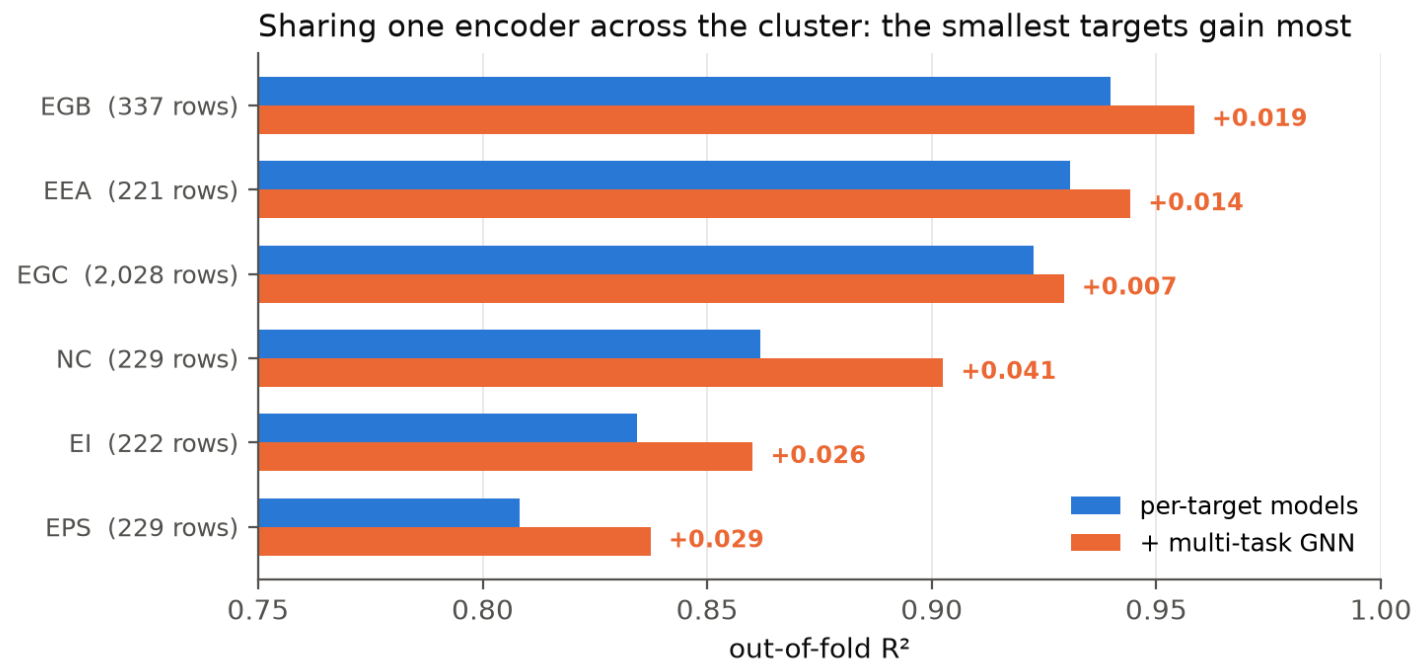
One encoder for the physics they share, one model per target for what they don't



electronic cluster only (egc, egb, ei, eea, eps, nc) — tg is thermal and trains alone

The shared encoder pays where data is thinnest

Gains scale inversely with training rows — exactly what transfer should look like.



Multi-task over the cluster: +0.0157 on the official metric.

+0.041

refractive index
229 rows

+0.029

dielectric constant
229 rows

+0.007

chain band gap
2,028 rows

Results

Out-of-fold, blend weights fitted by nested cross-validation so they never see the rows they score.

Target	Rows	LightGBM	GNN	Multi-task	Blend
EGC	2,028	0.908	0.913	0.927	0.930
EGB	337	0.902	0.940	0.958	0.959
EI	222	0.798	0.826	0.866	0.864
EEA	221	0.891	0.930	0.942	0.945
EPS	229	0.746	0.805	0.834	0.837
NC	229	0.810	0.860	0.906	0.903
TG	4,143	0.913	0.905	—	0.920

0.9139

official metric, mean R^2 over 7 targets

0.537

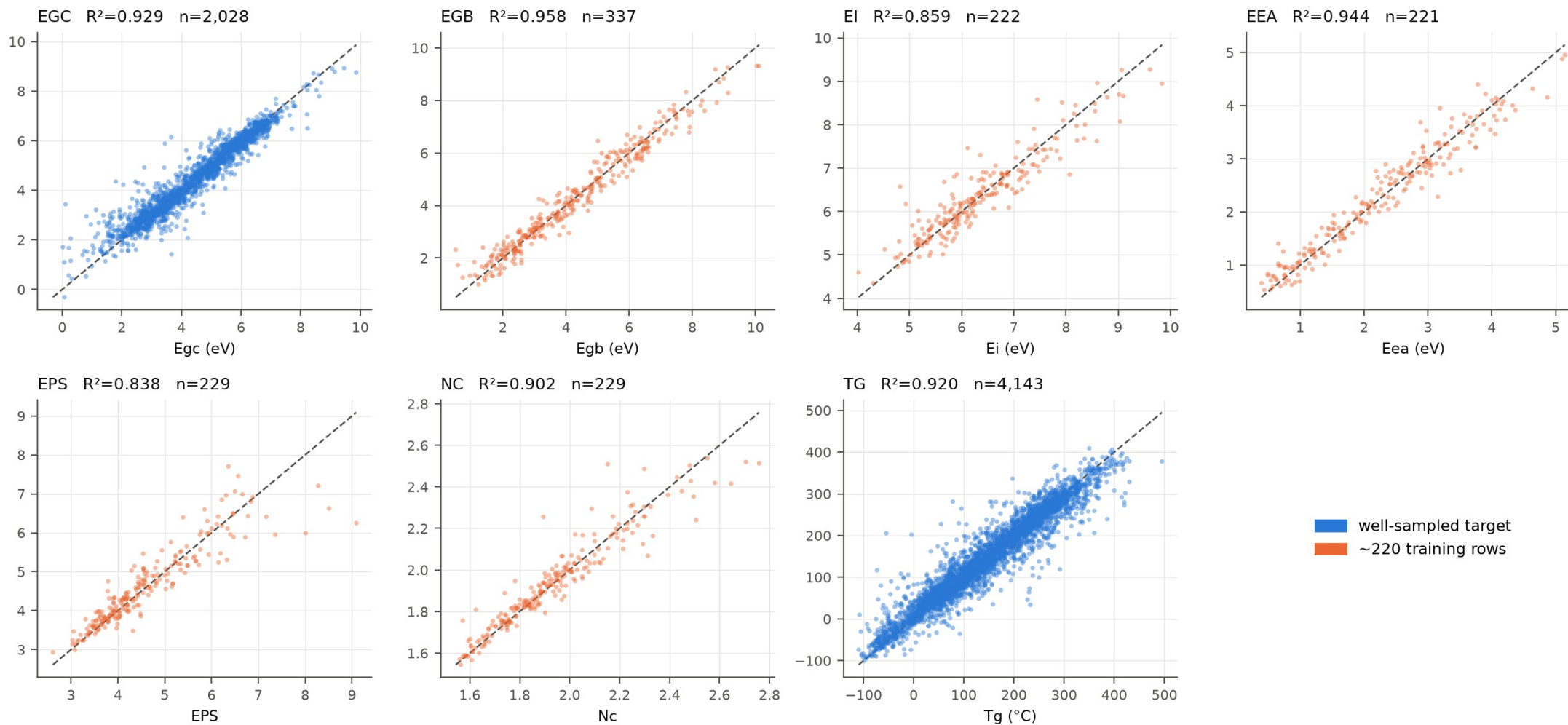
organizer Ridge baseline

Progression across submissions: 0.889 per-target → 0.904 adding the shared encoder → 0.908 seed-averaged → 0.914 at ten folds

Predictions against ground truth

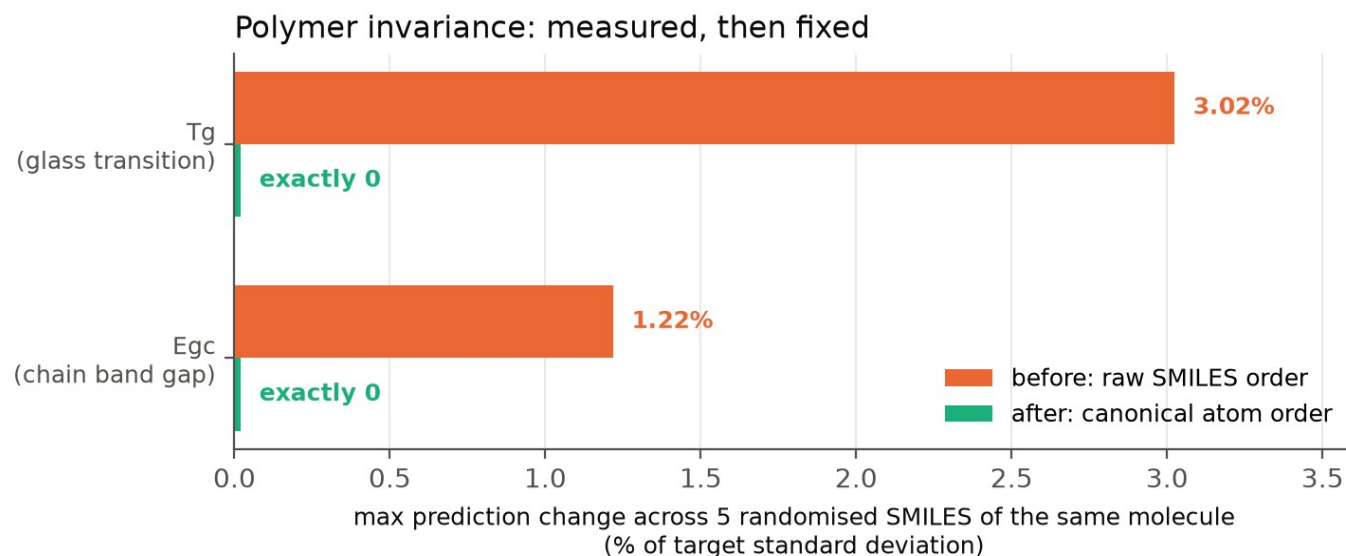
All seven targets, out-of-fold.

Out-of-fold predictions vs ground truth, all seven targets (final blend)



Polymer invariance: we measured it, then fixed it

A model that answers differently for two spellings of the same molecule cannot be trusted.



Found

predictions moved up to 3.02% of target sd across randomised SMILES

Cause

atom ordering — tied ring paths, Gasteiger charges, lpc rounding

Why it mattered

65% of the supplied SMILES are not canonical

Fix

re-parse from canonical SMILES, drop lpc. Cost: 0.0001

deviation after the fix: exactly 0

What did not work

Nineteen experiments across the competition. Four survived.

PI1M pretraining

-0.0038

Masked-atom pretraining on 100k unlabeled repeat units reached 99% reconstruction accuracy. The representation still did not beat supervised training on 3,266 labelled rows.

Multi-task including Tg

-0.0230

Tested in Round 2. Tg and Egc share no physics, so one encoder serving both diluted each. Same technique, opposite result — which is why we measured the correlations first this time.

Multi-task LightGBM

+0.0004

Too weak alone (R^2 0.79) to earn blend weight.

What changed from Round 2 to Round 3

The same technique, applied where the physics justified it.

Round 2 • two targets

- **Polymer backbone features**

18 scalars from the chain between the two attachment points. Worth +0.0056, more than 1,024 Avalon bits.

- **Message-passing GNN**

Reads topology directly; errors decorrelated from the trees. Worth +0.0053.

- **Multi-task across Tg and Egc**

Rejected: -0.0230.

Round 3 • seven targets

- **Measure the targets first**

Correlation structure decided the architecture before any model ran.

- **Multi-task across the electronic cluster**

Same technique that failed in Round 2, now +0.0157 — because these six targets are one physical quantity.

- **Invariance treated as a measurement**

Not an assumption. Found a 3% defect and removed it.

Model particulars

Everything runs from scratch inside one Kaggle notebook. No pretrained weights, no uploaded artifacts.

387k

parameters
multi-task encoder

10×3

folds $\times$ seeds
multi-task

3h 20m

training
Kaggle CPU, no GPU

<1 ms

inference
per molecule

Reproducibility. Single-threaded with per-fold seeds — parallel `index_add_` in message passing is otherwise non-deterministic, and we measured two identical runs differing by 0.18 per prediction before pinning it. Kaggle reproduced our local out-of-fold score within 0.0035 on every run of the competition.

What we would tell a materials scientist

Electronic

Band gap falls as conjugation runs along the backbone lengthen and as attachment atoms turn sp^2 — conjugation then propagates across repeat units instead of stopping at the unit boundary. Dimer conjugated-bond fraction correlates -0.76 with chain band gap.

Optical

Refractive index and dielectric constant rise with the same polarisable, aromatic-rich backbones that narrow the gap ($+0.92$ with each other, -0.85 with band gap). The absorption edge follows at $\approx 1240/E_{gc}$ nm. Side-chain bulk blue-shifts and solubilises.